LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Alighting gear, basic design requirements, types of undercarriage and functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Alighting gear, basic design requirements, types of undercarriage and functions
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Landing Gear Diagrams; Safety: Do not touch hydraulic lines; Instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear?
(b) Name two basic types of landing gear configurations.
(c) Why is shock absorption necessary during landing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements for landing gear • Load bearing and structural strength • Shock absorption mechanisms • Brake system integration • Wheel and tire specifications	L & Ds	PPT, IB	15 min
(b)	• Types of undercarriage: Conventional vs Tail-dragger • Tricycle configuration advantages • Retractable vs Fixed gear systems • Steering mechanisms and nose wheel • Main gear strut construction details	L & Ds	PPT, PDF	20 min
(c)	• Functions of oleo-pneumatic struts • Anti-squat mechanism operation • Ground handling and taxiing procedures • Landing gear door operation • Summary of gear construction elements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the oleo-pneumatic strut in absorbing landing impact.
(b) Differentiate between conventional and tricycle landing gear configurations.

12. Home Assignment and Follow Up	Prepare a brief note on the hydraulic circuit components involved in landing gear retraction to be discussed in LP 11.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety and emergency devices for retraction and extension, typical u/c circuits,	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min